VISHWAKARMA AWARDS
A Maker Bhavan Foundation Initiative

MAKER BHAVAN FOUNDATION

ARKA AI

VISHWAKARMA AWARDS 2026-27

VK26-IITH-SE-0021

Intelligent Edge-AI Monitoring & Performance Optimization for Solar Power Plants

THEME Sustainable Energy

Sub-theme: Renewable Energy Harvesting & Generation Systems

JUDGES: REFER TO THE REPOSITORY FOR IN-DEPTH DOCUMENTATION AND THE PROTOTYPE VIDEO.

https://github.com/DAnuraag/VK26-IITH-SE-0021

README documentation and prototype video will be added before judging.

Problem Context

Solar monitoring across an onsite embedded platform, website and companion app.

PROJECT OVERVIEW

ARKA AI

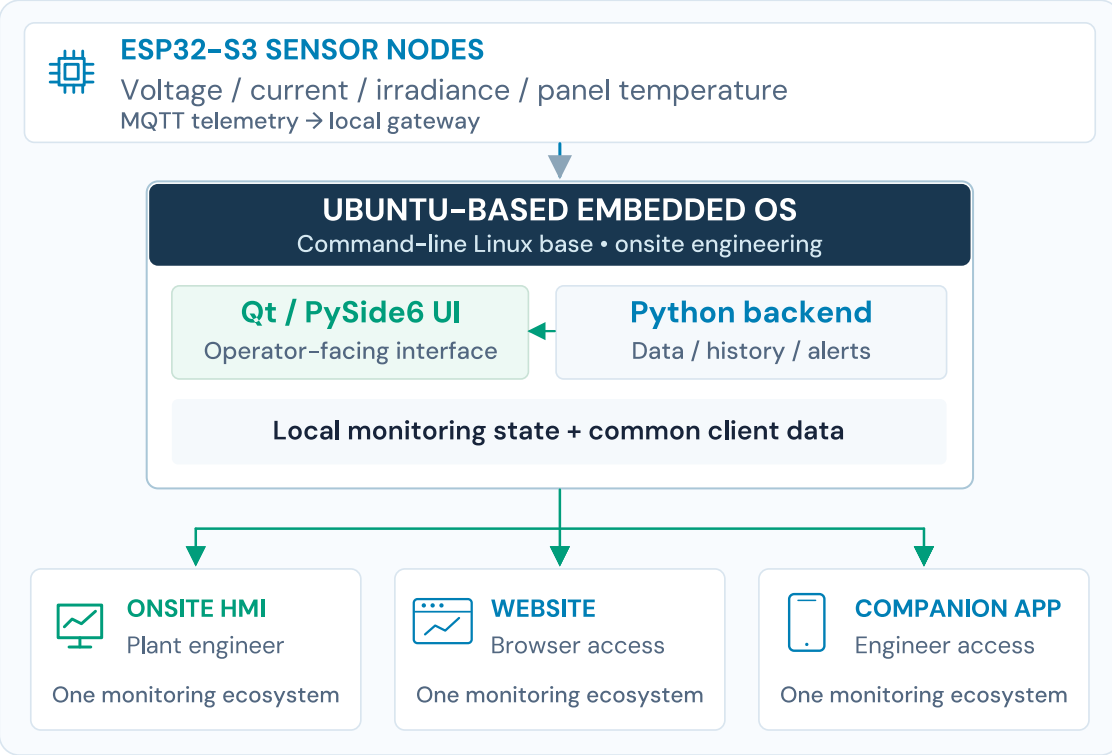
Solar sensing and edge analysis with an Ubuntu-based onsite system, Qt UI, Python backend, website and companion app.

Application ID: VK26-IITH-SE-0021

Track / theme: Sustainable Energy

The operational pain point

- Fragmented readings can hide zone underperformance.
- Changing sunlight and temperature complicate diagnosis.
- Delayed visibility pushes teams toward reactive maintenance.



For onsite engineers, plant operators, maintenance teams and renewable-energy asset owners.

One ecosystem: electrical + environmental context, local intelligence and connected engineering access.

Basis: Stage 1 project submission, §§2–8; Ubuntu / Qt / Python, website and app — implementation status.

Problem Context

2/2 · DOCUMENTED + PROPOSED

A retrofit decision-support layer—not a replacement for inverter controls or certified protection.

PLANT OPERATOR

See output and deviations through the website / app.

MAINTENANCE TEAM

Use the onsite Qt system; inspect and record outcomes.

ASSET OWNER

Improve visibility for informed maintenance planning.

WORKFLOW / ALTERNATIVE	POSSIBLE GAP — NOT UNIVERSAL	ARKA AI DESIGN RESPONSE
Periodic manual logs	Electrical and weather records may be disconnected.	Align V, I, irradiance and temperature in one time series.
Existing inverter / OEM telemetry	Useful data may already exist; integration varies by plant.	Use approved read-only data where available; avoid duplicate controls.
Cloud-only workflow, where used	A WAN outage can interrupt the remote view.	Keep gateway storage, analytics and the HMI on the local network.
ARKA AI ecosystem	Onsite and remote users need consistent monitoring context.	Ubuntu + Qt / Python, website and app share monitoring data and event context.

One monitoring ecosystem: local Qt access for engineers, website / app visibility and a common maintenance workflow.

No exclusivity or “first-ever” claim is made. Comparative gaps are deployment-dependent, not claims about every competing product.

User Journey

BEFORE: the physical deviation appears before a useful maintenance signal.

01 **Manual readings**
Periodic plant inspection rounds

02 **Scattered records**
Electrical and weather readings apart

03 **No context**
Actual output is not compared with a baseline

04 **Delayed flag**
Deviation waits for the next inspection

05 **Reactive dispatch**
Maintenance begins after a late signal

Visibility gap before an actionable signal

4 h
ASSUMED INSPECTION INTERVAL

2 h
MEAN WAIT UNTIL THE NEXT ROUND

1.00 kWh
UNOBSERVED SHORTFALL BEFORE FLAG

Illustrative plant-zone scenario: 5 kW × 10% shortfall × 2 h average detection wait = 1.00 kWh.

Assumptions: uniform fault onset between 4-hour rounds; constant 10% shortfall. This is not measured plant data.

The 5 kW zone is a hypothetical operational example; the proposed hardware demonstration remains 12–24 V DC, ≤5 A.

User Journey

AFTER: sensing → analysis → Qt / website / app visibility → inspection → recorded outcome.

SENSE
V / I / G / T
01

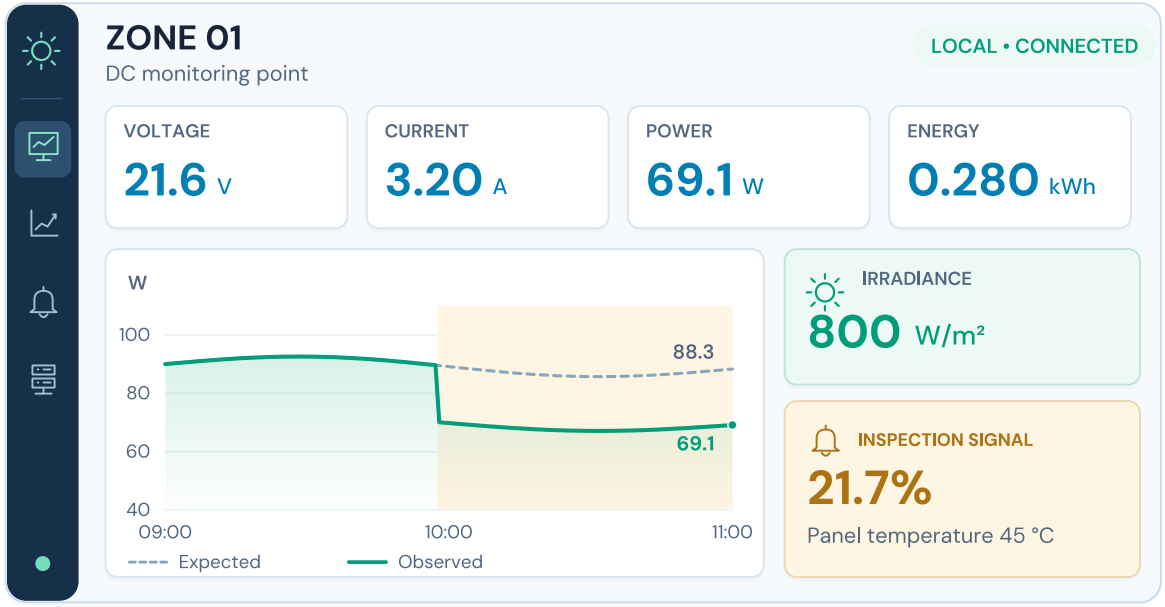
TRANSPORT
ESP32-S3 → MQTT
02

ANALYSE
Python + edge logic
03

VIEW
Qt • website • app
04

INSPECT
Engineer decision
05

RECORD
History + outcome
06



≤3.5 min
TARGET ALERT LATENCY: 180 s PERSISTENCE + 30 s MARGIN

97.1%
MODELED REDUCTION IN DETECTION WAIT VS 2 h

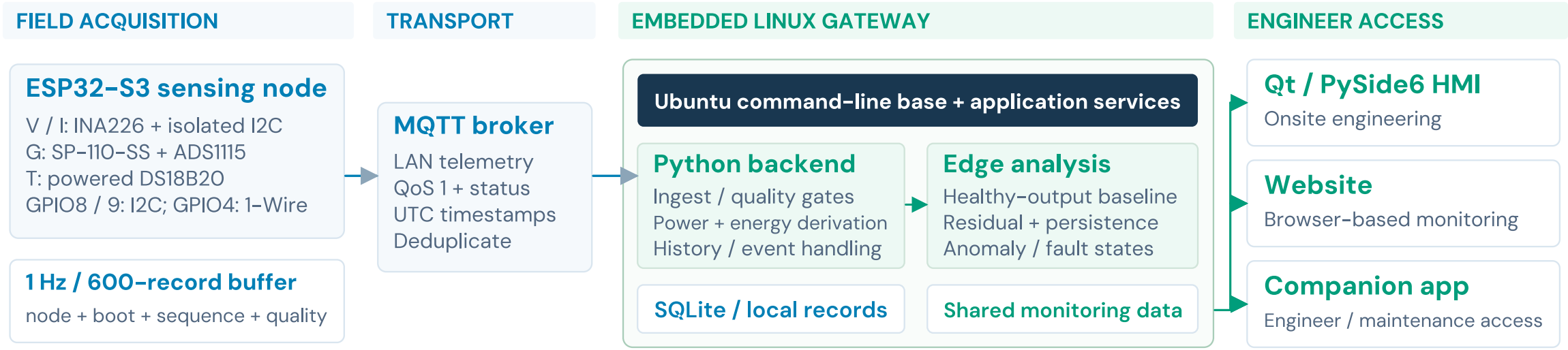
0.029 kWh
MODELED UNOBSERVED SHORTFALL IN THE SAME SCENARIO

Interface illustration • example values; software captures to accompany the README.

Same hypothetical 5 kW / 10% scenario as slide 4. Faster detection is modeled—not measured yield gain or guaranteed energy recovery.

System Architecture & Specs

Ubuntu-based embedded OS + Qt / Python backend + website + app, linked to one sensing workflow.



SYSTEM CONTRACT Shared site / node identity • timestamp • quality • event state across all clients

Local operation is independent of continuous WAN access; remote clients require an approved network path.

Logical ecosystem map • Ubuntu / Qt / Python, website and app form the software baseline; deployment details will accompany the README.

ONSITE EMBEDDED OS

Ubuntu CLI base + Qt UI + Python backend for onsite engineers.

WEBSITE + APP

Companion access to monitoring data, events and history.

SHARED WORKFLOW

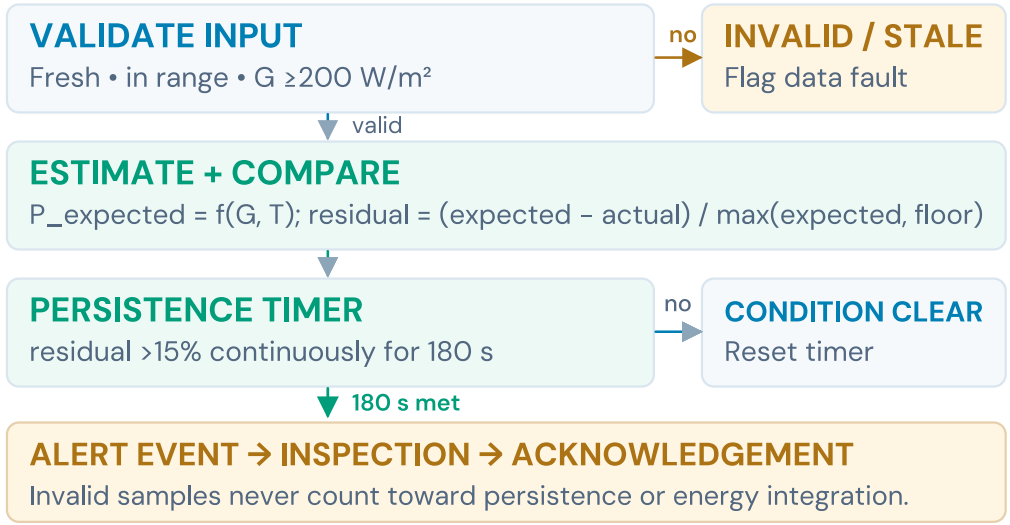
Common node, time, quality and alert context; human-led inspection.

Next: optional node-battery voltage, approved inverter / SCADA inputs and validated SOC / field-access workflows.

System Architecture & Specs

Testable operating targets; explainable anomaly logic before more complex models.

PARAMETER	TARGET — NOT ACHIEVED	IMPLEMENTATION
Pilot envelope	12–24 V DC; ≤5 A; one instrumented zone.	INA226 + 10 mohm shunt; isolated I2C.
Metrology	V / I error ≤1% above 0.5 A; power ≤2%, after calibration.	DMM / reference-meter comparison.
Context sensing	Irradiance 0–1,500 W/m²; panel temperature -10 to 85 °C.	SP-110-SS + ADS1115; DS18B20.
Timing / freshness	1 Hz publish; HMI data age ≤3 s; mark stale after 5 s.	Sequence + time; shared freshness / quality state.
Local retention	30 days raw data; 10-node scale test; 600-record node buffer.	64 GB endurance storage + SQLite.
Anomaly gate	Target precision ≥90%, recall ≥85%; qualifying alert ≤210 s.	Chronological held-out controlled events.



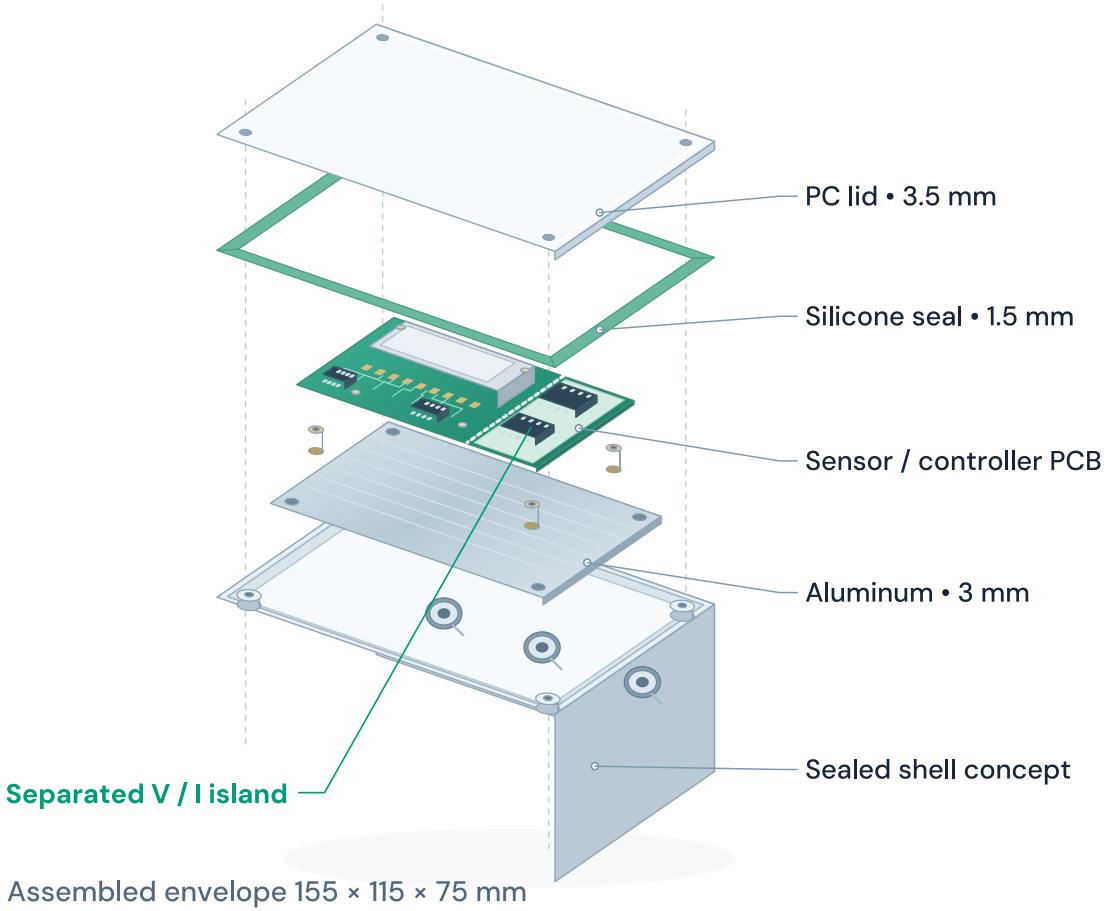
residual = (P_expected - P_actual) / max(P_expected, P_floor)

- Healthy baseline: expected output from irradiance + temperature.
- Gate low light (<200 W/m²) and fast cloud changes.
- Precision: correct-alert share. Recall: detected-anomaly share.

All figures are targets; testing is pending. Failure-life prediction is future work; device resolution is not system accuracy.

CAD Design & Mechanical Specs

A serviceable enclosure concept for separated sensing, controller and power paths.



Material and packaging rationale

UV-STABILIZED POLYCARBONATE

Proposed 3 mm shell and 3.5 mm lid: outdoor durability, impact resistance and accessible servicing.

ALUMINUM + SILICONE

125 × 85 × 3 mm backplate for rigidity / heat spreading; 1.5 mm gasket for the sealing concept.

LAYOUT / ASSEMBLY

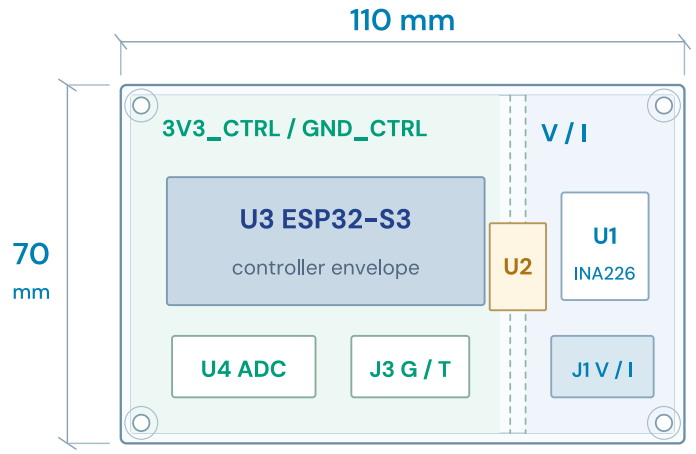
110 × 70 mm PCB envelope, M3 mounting points, strain-relieved cable entries and a separate V / I island. Routing and gland drilling need release review.

IP65 is a design target—not a tested system rating.

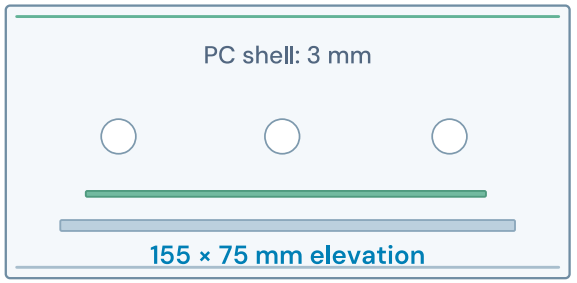
Sensor-node packaging concept; Ubuntu engineer-terminal hardware is separate. Native CAD / EDA release review remains necessary.

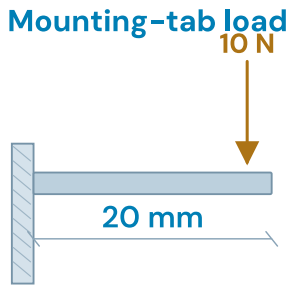
CAD Design & Mechanical Specs

Dimensioned PCB allocation, enclosure elevation and an explicit mechanical loading assumption.



4 × 3.2 mm clearance holes • M3 standoffs





b = 20 mm • t = 3 mm

Barrier corridor and component placement are allocation concepts only. Final spacing / footprints require PCB + safety review.

No copper bridge

between ISO / CTRL grounds

Mounting-tab screening — not FEA

Assume a 10 N load on one 20 mm cantilever tab; width 20 mm, thickness 3 mm; E = 69 GPa.

Bending stress = 6 F L / (b t²) = 6.67 MPa

Tip deflection = F L³ / (3 E I) ≈ 0.009 mm

FEA / RELEASE GATE — PENDING

Model screw bosses, fillets and gland cut-outs; document supports and loads; refine the mesh until stress / displacement change is <5%. No FEA result or factor of safety is claimed.

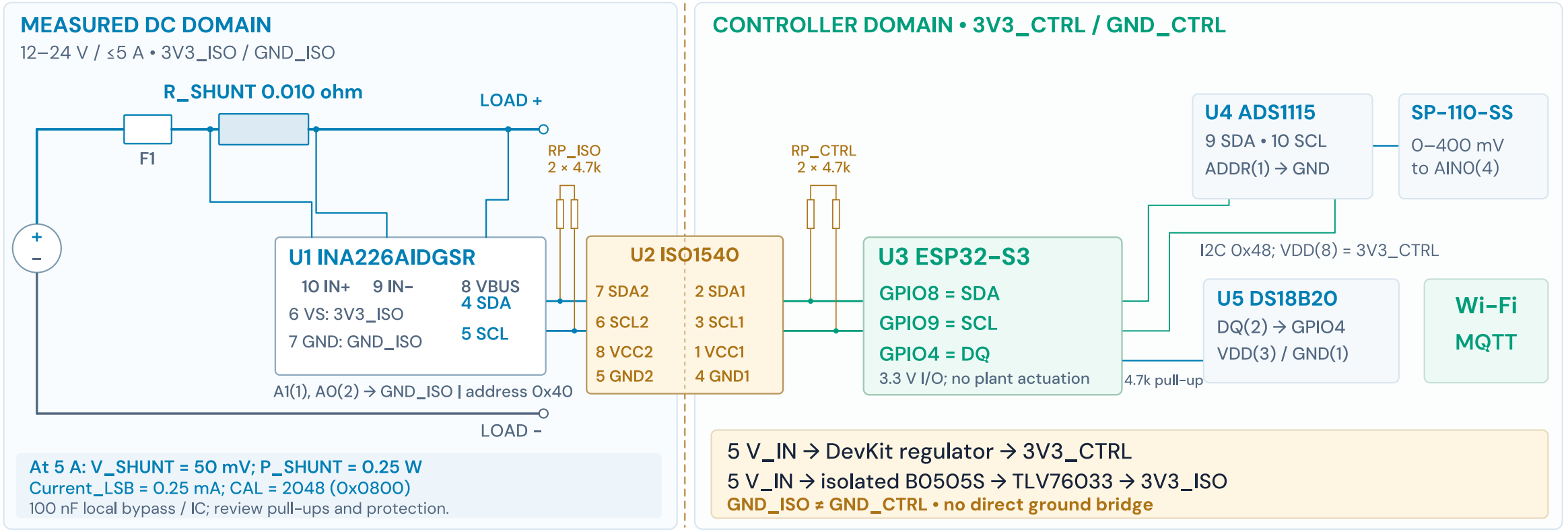
Dimensions in mm • views not to common scale • not released EDA / CAD.

Thermal target: 3 W node dissipation at 45 °C ambient → interior ≤65 °C; required effective thermal resistance ≤6.7 °C/W.

Analytical assumptions are proposed design inputs. They do not validate cut-outs, sealing, stress concentration, fatigue or outdoor safety.

Electronics Schematics & BOM

Pin-referenced connection concept: Kelvin sensing, isolated I2C, supply rails and controller I/O.



TI pin references checked: INA226 DGS p3 • ISO1540 D p4 • ADS1115 DGS p3. Named nets shown; protection / PCB release pending.

ESP32-S3
GPIO8 SDA · GPIO9 SCL · GPIO4 1-Wire

I2C / DECOUPLING
0x40 V / I · 0x48 ADC · 100 nF at each IC

ISOLATION RULE
VCC / GND each side; never bridge grounds.

Not fabrication-ready EDA; ERC / DRC and a detailed protection review are pending. Never attach this circuit to live plant PV strings.

Electronics Schematics & BOM

One-node prototype budget; the irradiance reference and gateway are shareable across zones.

COMPONENT	PROPOSED PART / SPECIFICATION	QTY	UNIT ₹	LINE ₹	SUPPLY / LEAD*
Controller	ESP32-S3-DevKitC-1-N8R8	1	1,300	1,300	India · 1–2 wk*
V / I measurement	INA226AIDGSR + 10 mohm / ≥1 W Kelvin shunt	1 set	650	650	India · 1–2 wk*
Isolation stage	ISO1540DR + B0505S-1WR3 + TLV76033DBZR	1 set	1,100	1,100	India · 1–2 wk*
Irradiance ADC	ADS1115IDGSR breakout	1	300	300	India · 1–2 wk*
Panel temperature	DS18B20 + cable / contact mount	1	250	250	India · 1–2 wk*
Irradiance reference	Apogee SP-110-SS pyranometer	1	32,000	32,000	Import · 3–6 wk*
Local gateway kit	Raspberry Pi 4B 4 GB + 64 GB card + PSU	1 set	6,800	6,800	India · 1–2 wk*
Node power supply	5 V / 3 A external PSU; GST18A05-P1J proposed	1	1,000	1,000	India · 1–2 wk*
Enclosure / mounting	PC case, gasket, glands + aluminum plate	1 set	1,600	1,600	Local · 1–2 wk*
PCB / protection	2-layer FR4 board + protective components	1 set	1,200	1,200	Local · 1–2 wk*
Harness / passives	Terminal blocks, cable, fuse and passives	1 set	500	500	Local · 1–2 wk*

₹46,700
ITEMIZED HARDWARE SUBTOTAL

₹56,040
WITH 20% PLANNING CONTINGENCY

₹7,900
NODE-ONLY ESTIMATE; SHARED ITEMS EXCLUDED

*All prices / lead times are planning assumptions, not vendor-verified. Excludes bench PSU/load, tax/shipping variance and field-rated HV interfaces.

Your Maker Journey

From the documented solar-monitoring architecture to an onsite + connected software ecosystem.

01 · PROJECT FOUNDATION

Solar monitoring foundation

- Defined V / I, irradiance and panel-temperature monitoring.
- ESP32-S3 → MQTT → local gateway, history and anomaly workflow.
- Established monitoring-first safety and modular zone expansion.

02 · EMBEDDED PLATFORM BUILT

Ubuntu + Qt + Python

- Built an Ubuntu command-line-based embedded OS environment.
- Qt / PySide6 UI with Python backend for onsite engineers.
- Local engineering interface for the monitoring workflow.

03 · CONNECTED CLIENTS BUILT

Website + companion app

- Built a website and app alongside the embedded platform.
- Connected the access surfaces as one monitoring ecosystem.
- Onsite and connected users share the plant-monitoring workflow.

Next handoff: README documentation + prototype video + build / screenshot evidence + cross-client and hardware verification.

Source basis: Stage 1 submission + software implementation status. Hardware and cross-client performance / safety remain verification gates.

Your Maker Journey

Repository handoff: explain the complete ecosystem and publish the prototype walkthrough.

JUDGES: PLEASE REFER TO OUR GITHUB REPOSITORY
FOR IN-DEPTH DOCUMENTATION AND THE PROTOTYPE VIDEO.
https://github.com/DAnuraag/VK26-IITH-SE-0021

Scan to open the repository

Canonical project link • QR verified

README.md documentation and the prototype-video link will be added.

Current basis: Stage 1 project submission + Ubuntu, Qt / Python, website and app implementation status.

README / REVIEW AREA	WHAT THE PROJECT DOCUMENTATION WILL CONTAIN
Embedded system	Ubuntu CLI base, Qt UI, Python backend, installation / boot and application-service workflow.
Website + app	Client connection flow, shared monitoring context, operating guide and actual interface captures.
Source + validation	Actual source / build files, lab photos, calibration and cross-client / offline / alert tests; safety scope.
Prototype video	External walkthrough of sensing → onsite Qt platform → website / app → alerts / inspection.

Write and upload the detailed README and real video link before judging; verify public access and preserve genuine build / test records.

Safety & Hazard Mitigation

DESIGN FMEA — PROPOSED

Monitor safely; fail visibly; retain certified electrical protection and human control.

FAILURE / HAZARD	PRIORITY	FAIL-SAFE / MITIGATION RESPONSE
Live-plant / overvoltage exposure	HIGH	Limit this prototype to 12–24 V. No live-string connection; field sensing requires rated interfaces + qualified review.
Shunt / harness short or overload	HIGH	Current-limited PSU, coordinated DC fuse, covered terminals and an accessible bench power disconnect.
Sensor drift / missing values	MED	Range + freshness checks, quality flags, reference calibration; never treat a stale value as a healthy reading.
LAN / gateway interruption	MED	Review Ubuntu service recovery; stale after 5 s, node replay and deduplication; local HMI remains independent of remote clients.
False alert / unauthorized access	HIGH	Human confirmation; read-only Qt / web / app scope; restrict shell, broker and client permissions; no plant switching.

IEC 61724-1

PV monitoring terminology / measurement framework.

IEC 61010-1

Measurement-equipment safety design reference.

IEC 60529 / IP65

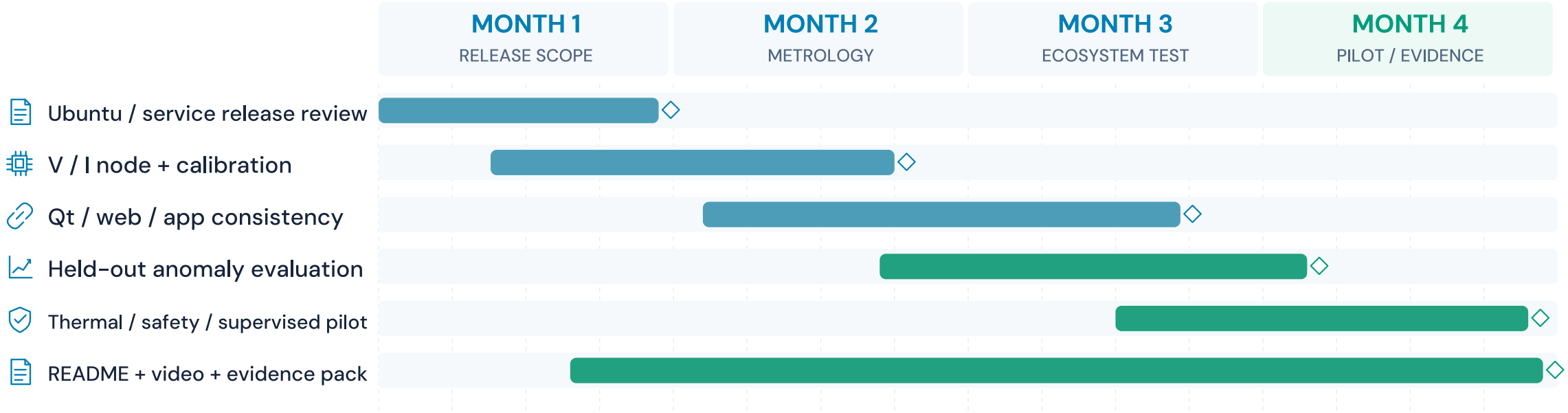
Enclosure target only; finished assembly testing pending.

Standards are design references—not a certification claim. Component isolation ratings do not certify the complete system for plant voltage.

Roadmap

FORWARD RELEASE PLAN

Validate and package the existing software ecosystem; complete sensing, safety and evidence gates.



Forward validation plan • Ubuntu / Qt / Python, website and app form the current software baseline.

M1 · RELEASE SCOPE

Review Ubuntu services, client interfaces and hardware safety scope.

M2 · METROLOGY

Calibrate V / I / power / energy and record reference uncertainty.

M3 · VERIFY ECOSYSTEM

Compare Qt / website / app states; test offline replay and alerts.

M4 · PILOT + HANDOFF

Supervised low-voltage pilot; README, video and real evidence pack.

Relative to approval; official Stage 3 dates still govern. The plan covers validation, deployment readiness, documentation and evidence.

Technical Challenges & Risks

ENGINEERING CONSTRAINTS

Measurement trust, environmental context, offline behavior and consistent Qt / website / app state.

BOTTLENECK	WHY IT MATTERS	EVIDENCE REQUIRED
R1 · Measurement envelope	INA226 is a low-voltage choice; shunt / wiring error matters at low current.	Range checks, calibration curves and a separate field-rated interface review.
R2 · Weather vs fault	Clouds, shading and temperature shifts can resemble electrical degradation.	Healthy baseline + event-level tests across normal / abnormal conditions.
R3 · Data continuity	Wi-Fi loss, clock drift and QoS 1 duplicates can corrupt apparent trends.	Source / receive time, sequence IDs, replay and deduplicated database checks.
R4 · Thermal / ingress	A sealed case plus sun can exceed the assumed 3 W / 45 °C thermal budget.	Real power / temperature logs, shaded mounting and finished-assembly ingress tests.
R5 · Cross-client integrity	Qt, website and app must agree on node identity, freshness, alert state and access permissions.	Same-event captures, controlled access / recovery tests and documented connection settings.

> Hard boundary: the prototype must prove trustworthy measurement and a local maintenance workflow before any higher-voltage plant integration.

These are anticipated validation risks, not passed tests. Procurement, calibration-reference availability and site approval remain dependencies.

Potential Solutions & Risk Mitigation

MITIGATION / ACCEPTANCE

Every risk has a primary action, a safe fallback and a measurable exit criterion.

RISK	PRIMARY SOLUTION	BACKUP / FALLBACK	RELEASE GATE — TARGET
R1	Kelvin shunt + multi-point V / I calibration.	Use an approved read-only reference / inverter interface for plant work.	≤1% V / I; ≤2% power; field safety approval.
R2	Quality gates + healthy-data expected-power residual.	Use explainable thresholds until a larger model improves held-out results.	Precision ≥90%; recall ≥85%; alert ≤210 s.
R3	QoS 1, unique sequence keys and 600-record replay.	Retain local gateway history; display unavailable / stale states explicitly.	600 s outage replay; no duplicate committed records.
R4	Sun shield, backplate heat path and protected cable entries.	Relocate electronics into a shaded, approved enclosure if thermal / sealing gates fail.	3 W / 45 °C → interior ≤65 °C; test final sealing.
R5	Use shared identity / event state; document and restrict client access.	Retain the onsite Qt HMI / local history when remote access is unavailable.	Same-event Qt / web / app consistency; access and recovery test logs.

Stage 3 goal: an auditable, calibrated, local-first monitoring prototype—not an unverified autonomous plant-control system.

Acceptance criteria are proposed. Use real test results to replace targets only after the tests have actually been completed.

References & Sources

Project submission, software implementation status and transparent quantitative assumptions.

Project documentation basis

ARKA AI — Stage 1 complete submission
Project documentation • 1 page.

- PDF §§2–4: problem, V / I + environment, MQTT / HMI and beneficiaries.
- PDF §§5–8: safety, modular expansion, innovation and user journey.
- Implemented software: Ubuntu CLI embedded OS, Qt / Python, website and app.

README documentation / video: forthcoming.
Official 20-slide template sequence retained.

DECLARED ASSUMPTION	CALCULATION / INTERPRETATION
Baseline model	4 h inspection interval ÷ 2 = 2 h mean detection wait.
Before exposure	5 kW × 10% × 2 h = 1.000 kWh unobserved shortfall.
After target	180 s persistence + 30 s margin = 210 s = 3.5 min.
After exposure	5 × 10% × 210 / 3,600 = 0.0292 kWh.
Wait reduction	1 - 210 / 7,200 = 97.08% (modeled, not measured).
Storage sizing	10 × 200 B × 1 Hz × 30 days = 5.184 GB, before overhead.
Mounting screen	F=10 N; L=20; b=20; t=3 mm → 6.67 MPa; 0.009 mm.
Prototype budget	₹46,700 subtotal + 20% contingency = ₹56,040.

JUDGE REFERENCE · https://github.com/DAnuraag/VK26-IITH-SE-0021 · README documentation / video forthcoming

All added dimensions, costs, thresholds and numeric goals are proposed planning inputs unless explicitly labeled as a source-device specification.

References & Sources

Manufacturer references guide component choices; standards guide the validation scope.

PRIMARY REFERENCE	USE IN THIS PROPOSAL	OPEN SOURCE LINK
Espressif · DevKitC-1	Controller GPIO / board pinout basis.	ESP32-S3 DevKitC-1 v1.1 [1]
TI · INA226	36 V bus monitor; ±81.92 mV shunt range.	ti.com/product/INA226 [1]
TI · ISO1540	Isolated I2C link; not complete-board certification.	ti.com/product/ISO1540 [1]
TI · ADS1115	Low-level pyranometer ADC / I2C interface.	ti.com/product/ADS1115 [1]
Analog Devices · DS18B20	Temperature interface; device accuracy has conditions.	analog.com · DS18B20 [1]
Apogee · SP-110-SS	0–400 mV output; nominal 5 W/m² per mV calibration.	apogeeinstruments.com · SP-110 [1]
Mornsun · B_S-1WR3	Isolated, unregulated DC/DC; use downstream regulation.	mornsun-power.com · datasheet [2]
TI · TLV76033	Fixed 3.3 V regulator for the isolated supply.	ti.com · TLV76033DBZR [1]
IEC 61724-1:2021	PV performance monitoring framework.	IEC Webstore · publication 65561 [1]
IEC 61010-1:2010	General measurement / control equipment safety reference.	IEC Webstore · publication 4279 [1]
IEC 60529 · consolidated	Enclosure IP classification / testing reference.	IEC Webstore · publication 2452 [1]

Ubuntu Linux documentation **Qt for Python / PySide6 documentation** **Python 3 documentation**

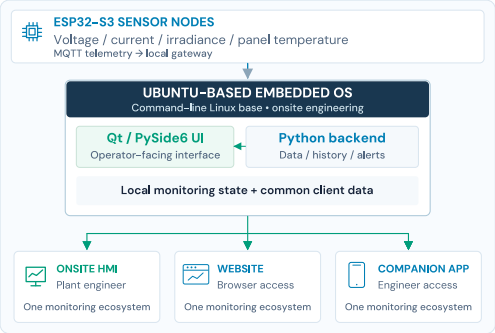
Manufacturer values are reference-device specifications, not verified prototype performance. No complete-system standards compliance is claimed.

Image Sources

FIGURE REFERENCES

Engineering figure register and reference basis for the ecosystem, measurement and packaging views.

Ecosystem & workflow



Ecosystem overview — slide 2

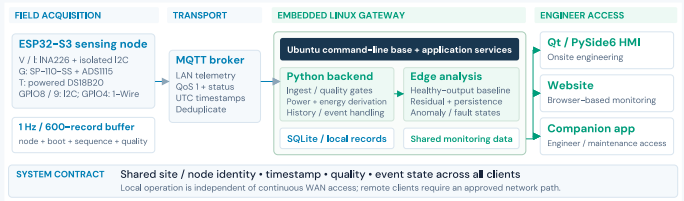
Before / after workflow — slides 4–5

Qt HMI reference layout — slide 5

Runtime / client architecture — slide 6

Basis: project PDF + team update

Measurement & logic



Anomaly state flow — slide 7

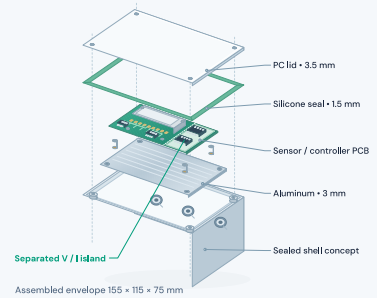
Electrical connections — slide 10

Verification roadmap — slide 15

Project repository QR — slide 13

Basis: component data sheets + design scope

Mechanical studies



Exploded enclosure — slide 8

Board / enclosure layout — slide 9

STEP assembly + part envelopes

Dimensions and analytical load case

Design concepts • release review pending

Official logos, banner and backgrounds: supplied Vishwakarma Stage 2 template. Interface / engineering illustrations are not lab photographs.

Figures explain architecture and design assumptions; actual software captures, fabrication and measured results require their underlying records.